

California Sovereign Infrastructure: the case in brief

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The accompaniment to the full proposal. Every figure here is the proposal's, labelled mid / critical, and the proposal carries its source and its case.

The problem

- California's residential load peaks after sunset and in summer, and the grid serves those hours with gas and imports at rising prices; a household's generation charge has reached about \$1,177 a year.
- The San Joaquin Valley's groundwater is overdrafted by 1.8–2.5 million acre-feet a year under a statute that requires the deficit to close; the coast has no new firm water source in service.
- Neither scarcity yields to demand reduction at scale, and every supply that can meet them is capital-intensive, first-of-a-kind, or both. This program addresses the two together, and prices both honestly.

The ask

- One state-owned program under one Authority, two severable projects, funded once by general-obligation bonds and never again by the treasury.
- Title I makes the electricity for three million households, 18.1 TWh a year, growing with population.
- Title II makes their water: 15–17 desalination modules, 0.75–0.83 million acre-feet a year, delivered year-round.
- Title III joins them so each helps the other and neither can drag the other down.
- Decision requested:** fund the studies and surveys now, \$213–463 million, and know within 2–4 years whether the rest is worth \$67–114 billion.

What it builds

- Helios-3.** Concentrating solar on three desert nodes — Mojave, Imperial, the Westside — with falling-particle receivers and supercritical-CO₂ turbines instead of salt and steam, so the same sun makes half again as much power (sun to socket 0.168 for the salt tower, 0.252 at mid).
- Daytime load goes straight from photovoltaics without touching a mirror; the mirror field is sized for the night; the particle store carries the evening peak.
- As sized at mid: 18.1 million m² of mirror on 14 towers, a 2,620 MWe block, 2,894 MW of PV. At critical: 30.9 million m², 24 towers, 2,620 MWe, 3,101 MW. The adopted route then grows the field by a quarter and the block by half to close the evening.
- Aqua-Sovereign.** Seawater reverse osmosis on retired coastal power plants, brine back through the permitted outfall at Ocean Plan concentration, no minerals sold, no zero-liquid-discharge.
- The joinder.** Title I's summer surplus lifts the water to an elevated reservoir; every evening of the year it comes down through pump-turbines to close Title I's peak. Water for the Westside and the coast, and the evening's last shortfall closed by the water on its way down.
- Together they leave **0.6% / 0.7%** of the load to the grid, against 13% / 13% with neither.

What it costs, once

	mid	critical
Title I, the plant and its risk register	\$32.7 B	\$55.4 B
Title I, closing the whole load with both levers	\$6.4 B	\$10.3 B
Title II, the water modules	\$28 B	\$48 B
program	\$67 B	\$114 B

- Every figure carries its contingency, 15% at mid and 30% at critical, and a one-year debt-service reserve.
- Every study and survey is the first line, \$213 / 463 million, priced and scheduled before a mirror is ordered.
- A salt-tower plant of this size costs \$46.1–71.5 B on the built record and does not pay for itself; this program builds a different plant and prices it from the record, not from a promise.

What a household pays

- Today: **\$1,177** a year to the utility for generation.
- On this program while the bonds are paid: **\$1,036** at mid, **\$1,682** at critical (\$172 / 279 per MWh).
- After the bonds retire: **\$190 / 263** a year, the running cost alone, for as long as the plant stands.
- 81% / 85% of the plant's bill is the build. Once built, the state owns a supply whose running cost is a fraction of today's bill.
- Water: **\$3,044 / 4,830** an acre-foot at cost recovery, in the band Carlsbad delivers (\$2,700–2,900), on its own rate and account.
- No further public money after the build: the bill carries debt service and O&M, the treasury carries nothing.

When

milestone	mid	critical
studies and surveys begin	2027	2027
first electricity, from the PV field at the first node	2031	2033
pilot receiver on sun, graded against a pass mark fixed in advance	2031–2033	2033–2035
first 100 MWe Helios-3 module	2036	2038
fleet complete, whole load served	2041	2043

- The realistic column is critical. The two-year slide is one item, the four-year federal environmental statement on the Mojave node, which is why the studies go first.
- Only a pass at the pilot orders the first module. A fail records the nitrate-salt plant on the same field as the route, already priced.

Why it holds up

- Every number comes from a calculation that can be re-run, printed at a middle case and at a critical case where every uncertain constant sits at its adverse end. The design is held to critical; mid is margin.
- The plant is sized to a requirement, three million households growing $1.35\text{--}1.81\times$ over the term, and the requirement is met before a kilowatt-hour is sold outside the state.
- It takes no developer's margin and no state subsidy after the build, and keeps the federal storage credit where statute allows.
- Every technology in it is on a register of 49 published studies and plants, each with a status and a source, and a recommended further study on a dated ladder.
- It mines nothing new, consumes no cooling water, and avoids 7.24 / 6.34 million tonnes of CO₂ a year with the load closed.
- About 21,054 construction jobs at peak and 1,901 / 1,214 permanent on the nodes, plus 601 / 664 at the water modules.

Where

- Title I: the Mojave (Kramer Junction), Imperial (Desert Center) and the Westside. At mid the Westside ties the Mojave for first on land, nexus and grid; at critical the Mojave leads on seismic margin. Land 45,361 / 66,339 acres, with fallowed Westside farmland covering a whole node.
- Title II: eight retired or retiring coastal plants screened on seven terms; the Oxnard plain (Ormond Beach, Mandalay) and Moss Landing lead; Huntington Beach carries its 2022 denial as a FAIL; no site is met on every term, so a site study is always owed.

What could stop it

- No plant of this kind has run at scale; the pilot comes first and the salt block stands behind it.
- The plant must deliver its modelled output, or the bill rises; no commercial salt tower has yet delivered its design.
- The coastal permits are the longest studies on the path and set the critical schedule.
- At the critical case a household pays more than today for one bond term. The proposal says so on every page, because a plan that hides its worst case is not one that can be built.

The decision

- Fund the studies now, for \$213–463 million, and know within 2–4 years whether the rest is worth \$67–114 billion.
- Nothing irreversible is bought before the receiver has passed its test.